Calculus and Linear Algebra

Taylor Applications
Sign, Limits and Local Behaviour

Lecture 25

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Today's plan

The first part of the lesson completes the Taylor-polynomial material from Lecture 24. The new material is deliberately compact:

1. Taylor expansions as **local models**;
2. limits from the first non-zero Taylor terms;
3. local sign and comparison of functions;
4. local extrema and inflection behaviour;
5. one short function-study application.

Time note

About 15 minutes of the lesson are reserved for the OPIS questionnaires.

From a polynomial approximation to a local model

Near x_0, Taylor's formula tells us that a smooth function behaves like a polynomial:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + R_n(x).$$

For local questions, the most important point is often not the whole polynomial but the **first non-zero term**.

Main idea

Close to x_0, the first non-zero Taylor term usually determines the scale, sign and shape of f.

Peano form: the language for local behaviour

A convenient local version of Taylor's formula is

$$f(x) = P_n(x) + o((x - x_0)^n), \qquad x \to x_0.$$

The notation

$$r(x) = o((x - x_0)^n)$$

means

$$\frac{r(x)}{(x - x_0)^n} \longrightarrow 0.$$

Interpretation

The remainder becomes negligible compared with $(x - x_0)^n$ as x approaches x_0.

Maclaurin expansions to keep available

As $x \to 0$:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3),$$

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5),$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4),$$

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3).$$

Strategy

Keep only as many terms as the problem requires. Expanding further is usually wasted work.

Order of a zero

Suppose

$$f(x_0) = f'(x_0) = \cdots = f^{(m-1)}(x_0) = 0, \qquad f^{(m)}(x_0) \neq 0.$$

Then

$$\boxed{f(x) = \frac{f^{(m)}(x_0)}{m!}(x - x_0)^m + o((x - x_0)^m)}$$

and therefore

$$f(x) \sim \frac{f^{(m)}(x_0)}{m!}(x - x_0)^m.$$

Meaning

The integer m is the order of the zero at x_0 and controls the local behaviour.

Limits: use only the first surviving terms

Exercise Compute

$$\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}.$$

Since

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3),$$

we obtain

$$e^x - 1 - x = \frac{x^2}{2} + \frac{x^3}{6} + o(x^3).$$

Hence

$$\boxed{\lim_{x\to 0} \frac{e^x - 1 - x}{x^2} = \frac{1}{2}.}$$

A higher-order cancellation

Exercise Compute

$$\lim_{x \to 0} \frac{\sin x - x}{x^3}.$$

Using

$$\sin x = x - \frac{x^3}{6} + o(x^3),$$

we get

$$\sin x - x = -\frac{x^3}{6} + o(x^3).$$

Therefore

$$\boxed{\lim_{x \to 0} \frac{\sin x - x}{x^3} = -\frac{1}{6}.}$$

Notice

The linear terms cancel, so the cubic term becomes decisive.

Another standard cancellation

Exercise Compute

$$\lim_{x\to 0} \frac{\ln(1+x)-x}{x^2}.$$

Because

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3),$$

we have

$$\ln(1+x) - x = -\frac{x^2}{2} + o(x^2).$$

Thus

$$\boxed{\lim_{x\to 0} \frac{\ln(1+x)-x}{x^2} = -\frac{1}{2}.}$$

How many Taylor terms do we need?

Consider

$$\lim_{x \to 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3}.$$

The denominator is of order x^3, so we must expand the numerator at least to order 3:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3).$$

After cancellation,

$$e^x - 1 - x - \frac{x^2}{2} = \frac{x^3}{6} + o(x^3),$$

so

$$\boxed{\text{limit} = \frac{1}{6}.}$$

A useful order comparison

Suppose, as $x \to x_0$,

$$f(x) \sim c(x - x_0)^m, \qquad g(x) \sim d(x - x_0)^n, \qquad c, d \neq 0.$$

Then

$$\frac{f(x)}{g(x)} \sim \frac{c}{d}(x - x_0)^{m-n}.$$

Hence:

$m > n$	ratio $\to 0$
$m = n$	ratio $\to c/d$
$m < n$	magnitude becomes unbounded

Practical use

Compare the first non-zero terms instead of manipulating the entire functions.

Exercise: mixed Taylor expansions

Exercise Compute

$$\lim_{x \to 0} \frac{e^x - 1 - x}{1 - \cos x}.$$

Use the local models

$$e^x - 1 - x \sim \frac{x^2}{2}, \qquad 1 - \cos x \sim \frac{x^2}{2}.$$

What is the limit?

Think before calculating

Both numerator and denominator vanish to the same order.

Exercise: solution

Since

$$e^x - 1 - x \sim \frac{x^2}{2}, \qquad 1 - \cos x \sim \frac{x^2}{2},$$

we obtain

$$\frac{e^x - 1 - x}{1 - \cos x} \longrightarrow \frac{1/2}{1/2}.$$

Therefore

$$\boxed{\lim_{x \to 0} \frac{e^x - 1 - x}{1 - \cos x} = 1.}$$

Alternative

L'Hôpital would also work, but Taylor shows immediately why the two quantities have the same local scale.

Local sign from the first non-zero term

Suppose

$$f(x) = c(x - x_0)^m + o((x - x_0)^m), \qquad c \neq 0.$$

Sufficiently close to x_0, the sign of f is the sign of

$$c(x - x_0)^m.$$

Therefore:

- if m is **even**, the sign is the same on both sides of x_0;
- if m is **odd**, the sign changes across x_0;
- the coefficient c determines which sign occurs.

Example: $e^x - 1 - x$ near the origin

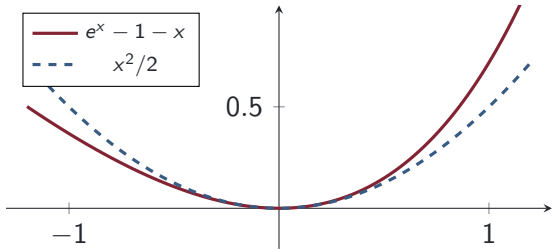
Taylor gives

$$e^x - 1 - x = \frac{x^2}{2} + o(x^2).$$

The first non-zero term has even order and positive coefficient. Hence

$$\boxed{e^x - 1 - x > 0}$$

for all sufficiently small $x \neq 0$.



Example: $\sin x - x$ near the origin

Taylor gives

$$\sin x - x = -\frac{x^3}{6} + o(x^3).$$

The first non-zero term has odd order, so the sign changes:

$$\sin x - x > 0 \quad (x < 0 \text{ near } 0),$$

$$\sin x - x < 0 \quad (x > 0 \text{ near } 0).$$

Geometric interpretation

Near the origin, the graph of $\sin x$ lies above its tangent $y = x$ on the left and below it on the right.

Taylor and local extrema

Assume

$$f'(x_0) = 0.$$

Look for the first derivative after f' that does not vanish:

$$f''(x_0),\ f^{(3)}(x_0), \dots, f^{(m)}(x_0).$$

Then locally

$$f(x) - f(x_0) \sim \frac{f^{(m)}(x_0)}{m!}(x - x_0)^m.$$

Higher-order derivative test

If m is even: positive coefficient $\Rightarrow$ local minimum; negative coefficient $\Rightarrow$ local maximum. If m is odd: no local extremum; the graph crosses its tangent level.

Examples beyond the second derivative test

At $x_0 = 0$:

$$f(x) = x^4 : \qquad f'(0) = f''(0) = f^{(3)}(0) = 0, \quad f^{(4)}(0) > 0.$$

Hence 0 is a local minimum. Similarly,

$$g(x) = -x^4$$

has a local maximum at 0, while

$$h(x) = x^3$$

has no local extremum: the first non-zero term has odd order.

Important

When $f''(x_0) = 0$, Taylor can still classify the point by looking at higher orders.

Local behaviour around a non-zero point

Taylor is not restricted to $x_0 = 0$. Consider

$$f(x) = \ln x - x + 1$$

near $x_0 = 1$. Set

$$t = x - 1.$$

Then $x = 1 + t$ and

$$f(x) = \ln(1 + t) - t.$$

Since

$$\ln(1 + t) = t - \frac{t^2}{2} + \frac{t^3}{3} + o(t^3),$$

we get

$$\boxed{f(x) = -\frac{(x-1)^2}{2} + o((x-1)^2).}$$

Thus $x = 1$ is locally a maximum and $f(x) < 0$ near 1, except at $x = 1$.

Function-study corner

Exercise Analyse the essential features of

$$f(x) = \ln x - x + 1.$$

Determine:

- domain and intercepts;
- limits at the boundaries of the domain;
- monotonicity and extrema;
- sign of the function;
- how the Taylor expansion around $x = 1$ confirms the local picture.

Function-study corner: solution

The domain is

$$D = (0, +\infty).$$

Moreover

$$f'(x) = \frac{1}{x} - 1 = \frac{1 - x}{x}.$$

Hence f increases on $(0, 1)$ and decreases on $(1, +\infty)$, with

$$f(1) = 0.$$

Also

$$\lim_{x \to 0^+} f(x) = -\infty, \qquad \lim_{x \to +\infty} f(x) = -\infty.$$

Therefore $(1, 0)$ is the global maximum and

$$\boxed{f(x) \leq 0 \quad \text{for all } x > 0.}$$

Taylor gives the same local conclusion because $f(x) \sim -\frac{1}{2}(x-1)^2$ near 1.

A sharper local-behaviour exercise

Exercise Let

$$g(x) = e^x - 1 - x - \frac{x^2}{2}.$$

Determine, near $x = 0$:

1. the first non-zero Taylor term;
2. the sign of $g(x)$ on the two sides of 0;
3. the value of

$$\lim_{x \to 0} \frac{g(x)}{x^3}.$$

Local-behaviour exercise: solution

From

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3),$$

we obtain

$$g(x) = \frac{x^3}{6} + o(x^3).$$

Thus the leading term has odd order and positive coefficient:

$$g(x) < 0 \quad \text{for } x < 0 \text{ near } 0,$$

$$g(x) > 0 \quad \text{for } x > 0 \text{ near } 0.$$

Finally,

$$\boxed{\lim_{x \to 0} \frac{g(x)}{x^3} = \frac{1}{6}.}$$

Taylor versus L'Hôpital

For many indeterminate limits, both methods work. Example:

$$\lim_{x \to 0} \frac{e^x - 1 - x}{x^2} = \frac{1}{2}.$$

L'Hôpital

Differentiate numerator and denominator, possibly more than once.

Taylor

Identify immediately the first non-zero term of the numerator.

Choice of method

Use the method that reveals the structure with the least unnecessary computation.

Common mistakes

- Expanding to too low an order when the first terms cancel.
- Expanding to a very high order when only one or two terms are needed.
- Using a Maclaurin expansion when the relevant point is $x_0 \neq 0$ without shifting the variable.
- Forgetting that the first **non-zero** term controls local sign.
- Assuming that $f''(x_0) = 0$ means "inflection point" without further analysis.
- Treating a local Taylor conclusion as a global statement without additional arguments.

What Taylor now allows us to do

Near a point x_0, Taylor expansions can answer three different questions:

How fast does a quantity vanish?

What is its sign close to x_0?

What local geometric behaviour does the graph have?

Core principle

Find the first non-zero term. Its coefficient and parity often contain almost all the local information we need.